Money and Power: The Impact of Tariff Structures on Electricity Consumption in Solar Microgrids in Africa

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***Abstract*—Electric utilities rely heavily on volumetric sales of electricity for revenue generation. Tariff designs have a direct impact on consumption patterns, hence revenue. Thus, it is important to design electricity tariffs such that they are efficient and fair. Here, we perform descriptive analysis on consumption data from four microgrids in Tanzania that have been subjected to time-of-use, block, and daily subscription tariff structures. We present two novel metrics, (1) Daytime-to-Total-Consumption Ratio (DTCR) and (2) Normalized Consuming Days (NCD), for assessing the electricity consumption patterns under each tariff. We find that a time-of-use tariff has minimal impact on electricity consumption time as it was only able to increase the DTCR of microgrids by 0.4% and 0.7%. We also find that microgrid customers consume more electricity under subscription-based tariffs than under consumption-based tariffs. In conclusion, this study provides empirical evidence on the inflexibility of nighttime electricity demand in rural African microgrids.**

***Index Terms*—microgrid, tariff, sub-Saharan Africa**

I. Introduction

The lack of universal access to electricity remains one of the most important challenges facing the world today and is one of the targets of the seventh Sustainable Development Goal (SDG7) [1]. This challenge is even greater in sub-Saharan Africa (SSA) where, according to the World Bank, as of 2019, only about 47% of the population (approximately 520 million people) had access to electricity [2]. While this marks a significant improvement from the less than 34% access rate recorded in the region a decade earlier, the International Energy Agency (IEA) estimates that the rates of new connections still falls short of what is required to realize (SDG7) [3]. Until recently, grid extension has been the main electricity access expansion strategy in most countries in SSA. However, over the last two decades, the cost of decentralized off-grid electricity technologies have fallen significantly, spurring the adoption of these technologies as alternatives to grid extension.

Despite this, the deployment of microgrids has not been substantial, largely because of the institutional structures and electrification policies which focus more effort and finances towards grid extension implemented by publicly owned utilities and leaving out the predominantly private sector led decentralized microgrids electrification mode. As development institutions continue to push for faster connection rates to achieve SDG7, there is an understanding of the need to accelerate private sector involvement in rural electrification efforts, particularly in the deployment of microgrids. However, the profit-driven nature of the private sector requires proving the business model of microgrids for rural electrification.

Setting aside grants and government subsidies, microgrids in SSA have two main sources of income: revenues from connection fees and from electricity sales. These two income streams are, however, not sufficient to cover the capital and operational costs of microgrids because microgrid developers rarely charge cost reflective fees [4]. For instance, the United States Agency for International Development (USAID) reports that microgrid connection fees range from $25 to $231 [5], despite the cost reflective connection fee ranging from $348 to $2,500 [6]. Interestingly, connection costs have been found to be a greater connection barrier than electricity consumption payments [7], leading to microgrid developers heavily subsidizing connection fees which further results in the sale of the electricity being the main source of income for microgrids.

In a bid to encourage participation by the microgrid sector in electrification efforts, most countries in SSA have adopted a flexible approach that allows microgrid developers to negotiate with regulators or target communities to determine the tariff that works for all parties, usually with a limit on the developer's allowable internal rate of return (IRR) — 18% for instance, in Kenya [4]. However, this approach is based on an assumed electricity demand and an expressed willingness to pay for electricity estimated from characteristics of a community based on pre-electrification surveys. While the accuracy of this method has been called into question in a recent study by Allee *et al.* [8], an aspect that the approach does not consider is how electricity demand varies with tariff designs. Overestimating demand for a given tariff level can impact the economics of the microgrid project while underestimating it can affect the reliability and availability of the microgrid, thereby motivating research aimed at understanding how tariffs

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influence electricity demand and consumption patterns.

Although there has been a lot of research devoted to electricity tariff design, most studies focus on national utility tariffs and on grid-connected microgrid tariffs, especially with the rise in the adoption of rooftop solar in the developed world [9]. Few studies have researched electricity tariff design for off-grid microgrids, particularly those in SSA. In a study of a run-off-the-river hydroelectric microgrid in Tanzania by Ngowi *et al.* [10], it was found that switching from a metered (consumption-based) tariff to a system with uniform monthly electricity bill regardless of consumption (subscription-based tariff) increased both electricity consumption and monthly revenues. However, the authors noted that the subscription tariff model may have resulted in overloading the microgrid since electricity users had little incentive to conserve energy. While this tariff model may work for small-hydro systems, it may not be financially feasible for the more commonly deployed hybrid diesel/PV/battery microgrid types in the region due to the variability and intermittency of solar resource and high diesel prices. Many of the studies discussing tariffs in microgrids in SSA tend to focus on the supply side of the relationship, that is, the impact of electricity demand on the unit cost of generating electricity, usually with the aim of highlighting the benefits of introducing productive-use-of-energy appliances into microgrids [11].

In this study, we use descriptive analysis to assess the impact of various tariff structures on electricity consumption patterns in off-grid microgrids. We provide two specific contributions: (1) provision of empirical evidence of the reaction of electricity users in decentralized microgrids to various tariff structures, and (2) assessment of the effectiveness of the tariffs in changing electricity consumption patterns and in achieving the objective that motivated their design. By shedding more light on the energy use patterns of microgrid customers under different tariff regimes, this study aims to reduce the uncertainty in the business models of energy project developers as regards the relationship between tariff structure and demand patterns, thereby attracting the much needed private finance into the microgrid sector of SSA.

II. Data and Tariff Structures

A. Data

Hourly-resolved consumption time-series data spanning five years collected from four microgrids owned and operated by PowerGen Renewable Energy were analyzed in this study. In each of these microgrids, there has been at least two of the three commonly deployed tariff structures in SSA (that is, block, time-of-use and subscription tariffs).

B. Tariff Structures

Table I provides a summary of the different tariff structures that were analyzed. These tariff structures are variants of consumption-based (volumetric) tariffs and subscription-based (fixed) tariffs.

TABLE I
Tariff structures

Time-of-use (TOU) tariff		
Tariff name	*Time groups*	*Charge (TZS/kWh)*
TOU 1	4:01 PM - 9:59 AM (Peak) 10:00 AM - 4:00 PM (Off-peak)	3000 1500
TOU 2	4:01 PM - 9:59 AM (Peak) 10:00 AM - 4:00 PM (Off-peak)	5000 2500
Block tariff		
Tariff name	*Threshold (kWh)*	*Charge (TZS/kWh)*
Block 1	Less than 3 Greater than 3	4000 2500
Block 2*	Less than 1 Greater than 1	3000 500
Block 3	Less than 3 Greater than 3 and less than 6 Greater than 6	4000 3000 2000
Daily subscription tariff		
TZS500 or 1000 for limits of 5kWh/day or 10kWh/day respectively		

* includes fee of TZS1000/month.

1) Time Of Use Tariff (TOU): TOU tariffs are typically aimed at shifting demand from the peak times of the day, which usually occurs in the late evenings in most rural microgrids in SSA, to offpeak times [12]. The objectives of TOU tariff are to shift the peak demand to match the solar PV generation profile by encouraging customers to use more electricity during the daytime and to price electricity such that it is reflective of the cost of generation. With this tariff structure, electricity is 50% less expensive in the daytime, incentivizing electricity users to shift their nighttime demand to daytime. When properly implemented in a hybrid solar microgrid with flexible loads, ToU tariff can shift demand to the daytime, reducing generator run-time and battery cycling.

2) Block Tariff: Block tariffs aggregate microgrid customers into groups based on characteristics such as electricity consumption level, and power requirements. In the microgrids studied, block tariffs were used to implement price discrimination based on electricity consumption level, with the first few kilowatt-hours being the most expensive. This tariff design was aimed at increasing the revenue of the microgrid by motivating higher electricity consumption. This was achieved by reducing the marginal cost of electricity when consumption exceeds a pre-defined threshold. It should be noted that the block tariffs in this study are decreasing block tariffs which are unlike the increasing block tariffs common in main grids that aim to reduce electricity cost for low-consuming electricity users.

3) Daily Subscription: Under daily subscription tariffs, customers are charged a daily fee to use a limited amount of energy, which expires at the end of the day. In the microgrids under study, the energy caps were set at 5 kWh and 10 kWh with different corresponding subscription rates. A benefit of this tariff structure is that it makes revenues more predictable, since microgrid developers can expect to collect a fixed amount from each customer on days that they subscribe. Daily subscription tariffs were employed in response to regulatory restrictions that attempted to make the tariffs of microgrids at parity with that of the grid.

III. Methods

A. Impact assessment

To assess the impact of tariff structures, a "before and after" analysis was performed, which involves comparing the electricity usage pattern of each customer before and after each tariff change. We were limited to this type of analysis because the assessment is conducted ex-post (i.e., after the tariff changes have occurred) and because the switch over from one tariff structure to another was done simultaneously for all customers. We acknowledge that our analysis may be prone to biases that could have been mitigated by randomized controlled trials (RCT), which is the approach prescribed in literature for assessing the impact of interventions aimed at achieving behaviour change [13]. We attempt to isolate the effect of tariff change from other longer term drivers of change in demand such as seasonality, organic demand growth and arrival of the grid, by limiting our assessment to 12 weeks before and after each tariff change. We also ensure that the same customers are being tracked over time, that is, we excluded customers that only got connected after the tariff change. The assumption here is that the electricity consumption pattern would not have changed without the tariff change and that the 12 weeks period is enough time for the customers to adjust their behaviour in response to the tariffs.

B. Data processing and analysis

The hourly consumption data had missing data and this was addressed by calculating the average hourly consumption for every hour of the day for each customer during the periods before and after each tariff change as shown in Eq. (1) where x_{ijk} is the *i-th* consumption reading of connection *j* at hour *k* of the day, with *k* ranging from 1 to 24. *n* is therefore the number of times the connection has consumption reading for a given hour.

$$h_{jk} = \frac{1}{n} \sum_{i=1}^{n} x_{ijk} \quad (1)$$

$$d_j = \sum_{k=1}^{24} h_{jk} \quad (2)$$

The hourly consumption h_{jk} was then summed up to obtain the average daily consumption of each customer in each period (d_{jk}) as in Eq. (2). Other analysis and calculations performed include: calculating the daytime-to-total-consumption ratio (DTCR) for each tariff regime to assess the tariff's impact on the time of use of electricity, and computing the normalized consuming days (NCD) which for each tariff regime is the sum of the number of days in the 12 weeks period that the microgrid customers consume electricity. This metric was normalized by dividing by the maximum consuming days (that is, the product of the number of customers and the number of days in 12 weeks).

IV. Results and Discussion

There are six distinct tariff changes (cases 1 to 6) that were implemented in the microgrids under study as detailed in Table II. The table also includes descriptive analysis of the impacts of the three tariff structures. Figure 1 below compares the average monthly consumptions in the tariff regimes and also indicates if the means of the consumptions are different at a 95% confidence interval as determined with a paired t-test. The effectiveness of the tariffs in achieving their goals was evaluated before discussing the implications of our findings on policies and microgrid design and operations.

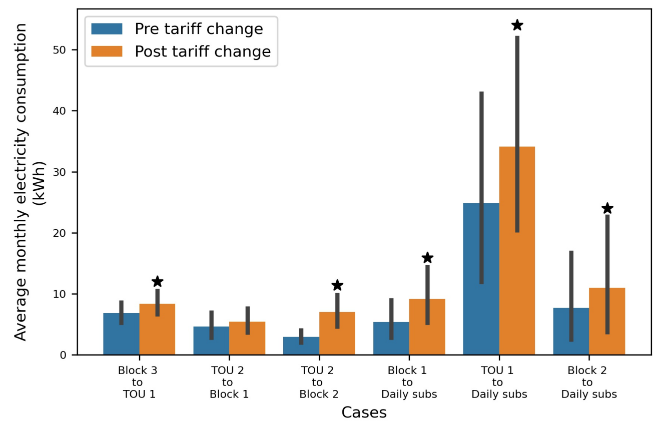

Fig. 1. Average monthly consumption before and after implementing tariff changes. Asterisks are placed on bars that are greater than the other at a 95% confidence interval (result of paired t-test). In case 2, we fail to reject the null hypothesis that the means of the consumption before and after the tariff changes are equal.

A. Effectiveness of tariff changes

1) Time-of-use tariff: The results of our analysis indicates that TOU tariff was not successful in motivating electricity users to shift their consumption to the daytime. In cases 1 and 2, TOU achieved small but positive changes in DTCR (change of 0.4% and 0.7%, respectively), as shown in Figure 2. In cases 3 and 5, the DTCR was actually lower with TOU than with block and daily subscription tariffs. We posit that this is because in case 3, the "post tariff" is block 2, in which after the first kWh, the unit cost of electricity reduces considerably. Also, in case 5, the "post tariff" is daily subscription which generally led to higher consumption and flatter load profiles than other tariffs since it offers little to no incentive to conserve energy. When not motivated to conserve energy, electricity users are more likely to leave their appliances on during the daytime leading to an increased daytime consumption which resulted in higher DTCR in the subscription-based tariffs in cases 3 and 5. In summary, TOU tariff did not significantly impact the time of electricity consumption in the microgrids.

2) Block tariff: Block tariffs aimed to incentivize electricity users to increase their consumption from a base level to a higher level. Two sites changed their tariff to block tariffs (case 2 and 3). We assess the effectiveness of the tariff in increasing consumption by counting the number of connections whose

TABLE II
CASES OF DIFFERENT TARIFF STRUCTURE CHANGES THAT WERE IMPLEMENTED

			Tariff Name		Avg. Consumption (kWh/month)		DTCR		NCD	
Case	Site	Customers	Before	After	Before	After	Before	After	Before	After
1	A	199	Block 3	TOU 1	6.9	8.4	23.4%	23.8%	0.68	0.95
2	B	54	TOU 2	Block 1	4.5	5.4	24.9%	24.2%	0.85	0.73
3	C	94	TOU 2	Block 2	3.0	6.9	22.4%	25.9%	0.87	0.92
4	D	200	Block 1	Daily Subscription	5.4	9.0	25.4%	36.1%	0.82	0.55
5	A	210	TOU 1	Daily Subscription	24.9	34.2	25.8%	34.80%	0.94	0.82
6	C	90	Block 2	Daily Subscription	7.8	11.1	21.9%	26.3%	0.58	0.53

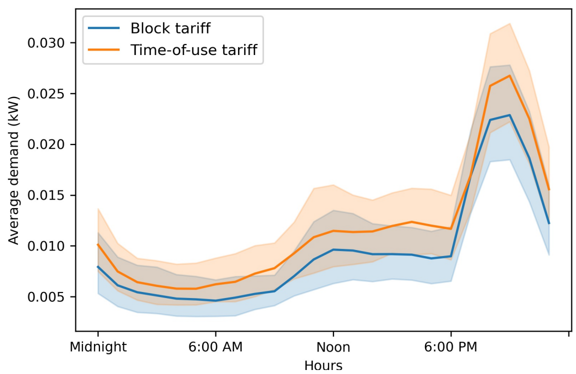

Fig. 2. Average daily load profiles in case 1 for time-of-use and block tariffs. Notice that there was no significant shift in the electricity demand from the nighttime to the daytime despite the 50% discount in the daytime. N.B. the translucent error bands around the lines shows the 95% confidence interval estimated with bootstrap resampling.

average consumption in the 12 weeks period was higher than the base level (3kWh in block 1 and 1kWh in block 2) and the result of this is presented in Table III. Block tariffs appears to be effective in increasing the share of connections consuming at consumption level 2 in case 2. However, in case 3, the opposite is observed. The structures of the block tariffs implemented provides a plausible explanation for this. In case 2, the block tariff only had a consumption component, while the block tariff in case 3 had both a consumption and a monthly subscription component. The results suggest that block tariff alone could be effective in increasing consumption levels and that the introduction of subscription-based block tariff may reduce average consumption as some electricity users, especially low consuming users, may not subscribe consistently. It should be noted, however, that a larger sample size will be required to validate these findings. Table III also shows the average monthly consumption of the customers assigned to the block levels before and after the tariff change was effected and the impact of the tariff on electricity expenditures. We consistently find that block tariff increased the consumption of level 2 consumers, as expected because the marginal cost of electricity at this level is lower. Despite this, this table also shows that block tariff generally reduced the average monthly expenditures of microgrid customers (and by extension, the microgrid revenues). The only time that block tariff results in an increase in expenditure is in case 3 and we suspect that this increase is driven by the subscription fee of 1000 TZS/month that this tariff includes.

3) Daily subscription tariff: Electricity users are expected to increase their electricity consumption with daily subscription tariffs relative to volumetric tariffs on days when they decide to subscribe because the marginal cost of consumption is zero up to the subscribed cap (which is generally much higher than consumption in other tariffs). In the three cases where this tariff was implemented, the average consumption increased regardless of the previous tariff with percentage increases ranging from 37% to 67%. However, the average monthly consumption when converted to average daily consumption is still quite far from the daily energy limits of 5kWh or 10kWh mentioned in Table I. We postulate that is because, even in days when customers subscribe, they may not have the high consuming appliances required to exhaust the amount of energy available to them. We also used the normalized consuming days (NCD) metric to assess the impact of subscription-based tariffs on the day-to-day consistency of electricity consumption of the microgrid customers. The hypothesis here is that with subscription-based tariffs, customers will consume electricity from the microgrid for a lower number of days since they may be unable to afford the subscription fee everyday. In this situation, customers may begin to seek alternative source of electricity or simply go without it. Table II shows that as expected, daily subscription-based tariff regimes had lower NCD in all three cases where it was implemented (cases 4, 5, and 6). The results suggest that microgrid customers consume electricity more consistently with consumption-based tariffs than with subscription-based tariffs as evidenced by the comparatively higher NCD in consumption-based tariffs.

Figure 3 shows that the daily subscription tariff resulted in significantly increased consumption at most hours of the day although daytime consumption increased more than nighttime consumption. This could be because daily subscription enables productive use of electricity which usually occurs in the daytime. We also observe that the consumption in the early mornings remains low even with daily subscription tariffs.

B. Policy recommendations & implications

The results described above have implications for both microgrid operators and electricity regulators. From a microgrid design and operations perspective, the minimal impact of the TOU tariffs on electricity consumption patterns highlight the inflexibility of the nighttime electricity consumption of microgrid customers. Solar-powered microgrids in the region should, therefore, include backup systems such as diesel

TABLE III
EFFECTIVENESS OF BLOCK TARIFF IN CHANGING CONSUMPTION LEVEL AND MONTHLY ELECTRICITY EXPENDITURE

	Changing Consumption Level								**Monthly Elecricity Expenditure**			
	Before block tariff				**After block tariff**				**Before block tariff**		**After block tariff**	
Case	Connection Count		Av. Monthly Cons. (kWh)		Connection Count		Av. Monthly Cons. (kWh)		Average Expenditure (TZS)		Average Expenditure (TZS)	
	Lv. 1	Lv. 2	Lv. 1	Lv. 2	Lv. 1	Lv. 2	Lv. 1	Lv. 2	Lv. 1	Lv. 2	Lv. 1	Lv. 2
2	31	23	1.72	7.59	27	27	1.29	9.61	7,771	32,860	5,171	28,520
3	22	72	0.42	3.68	42	52	0.55	9.0	1,962	16,360	2,645	7,994

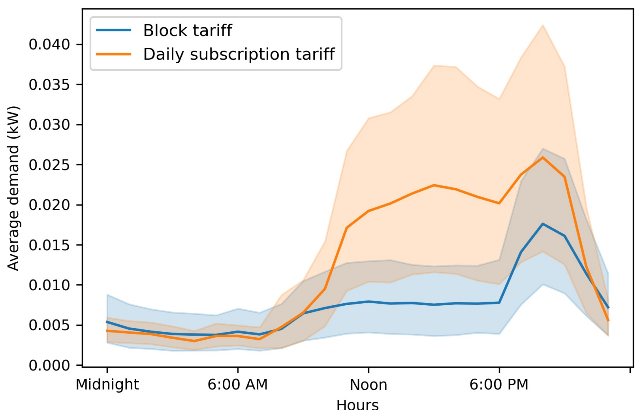

Fig. 3. Average daily load profiles in case 4 for the daily subscription and block tariffs. Notice the significant increase in daytime consumption. N.B. the translucent error bands around the lines shows the 95% confidence interval estimated with bootstrap resampling.

generators and battery systems to align the supply of solar energy with the demand for electricity. Furthermore, the NCD metric indicate that microgrid customers are more likely to consume electricity consistently with consumption-based tariffs than with subscription-based tariffs. Microgrid developers should, therefore, explore consumption-based tariffs when they aim to increase microgrid connection rates and discourage microgrid customers from seeking other sources of electricity such as solar home systems. This is largely driven by the fact that consumption-based tariffs have the payment flexibility advantage that aligns with the irregular nature of income in the mainly agrarian rural communities in SSA.

Lastly, this study also offers some insight to development institutions. While daily or monthly subscription tariffs has been shown in this work to increase the electricity demand of rural communities, we also observe that the amount of energy consumed are far from the imposed energy limits. This is indicative of the need for appliance financing and demand stimulation activities that will enable microgrid customers to increase their electricity consumption productively.

V. CONCLUSIONS & FUTURE WORK

This study assessed the impact of three tariff structures which are variants of consumption and subscription-based tariffs on electricity consumption patterns in four microgrids in Tanzania. Our results indicate that the nighttime demand in microgrids is generally inflexible and that offering 50% discounts to use electricity during the daytime has little impact on the nighttime demand. Future studies should test if increasing the discount in the off-peak periods can enable demand response with time-of-use tariff.

We also found that microgrid customers are less likely to use electricity on a daily basis with subscription-based tariffs which may be due to the nature of economic activities in the mainly agrarian communities that microgrids serve. More sophisticated statistical methods, especially those that help establish robust counterfactuals on which impacts will be measured, could shed further light on the statistical significance of our findings. Importantly, our descriptive analysis provides a basis on which hypotheses can be formed for future studies on consumer behavior and electricity tariffs in microgrids.

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